

MTH6134 Statistical Modelling II

Group Project: Data

Midland Premier League 2023/24

Overview

The group project for this module is based on data from the fictional 2023/24 Midland Premier League season. The datasets are described below. The full project brief, including the specific tasks your group will be asked to address, will be released later in the term.

In the meantime, you should use the PC sessions each week to explore these data using the statistical concepts and methods covered in that session. This will put you in a strong position when the full brief is released.

Groups will be allocated once registrations have settled. You can start exploring the data individually straight away; once your group is allocated, set up a shared workspace and keep your work together as described in the Working as a Group document.

The Data

Two datasets are available on QM+.

`league_players.csv` contains data on 620 players across all 24 clubs in the Midland Premier League: one row per player.

`ashford_matches.csv` contains data on all 46 league fixtures played by Ashford United during the season: one row per match.

2.1 Variable descriptions

Variable	Dataset	Description
<code>team</code>	Players	Club name
<code>position</code>	Players	Playing position: GK (goalkeeper), DEF (defender), MID (midfielder), or FWD (forward)
<code>age</code>	Players	Age in years
<code>minutes_played</code>	Players	Total minutes played during the season
<code>training_load</code>	Players	Weekly training intensity, scored 1 (low) to 5 (high)
<code>prev_injuries</code>	Players	Number of injuries in the three prior seasons
<code>bmi</code>	Players	Body mass index
<code>injured</code>	Players	1 if injured during the season, 0 otherwise
<code>home_away</code>	Matches	Home or Away
<code>goals_scored</code>	Matches	Goals scored by Ashford United
<code>goals_conceded</code>	Matches	Goals conceded by Ashford United
<code>opponent_rank</code>	Matches	Final league rank of the opponent

All data are simulated. The league, clubs, and players are entirely fictional.

Suggested weekly activities

Each week in the PC lab, spend some time with the data using the method covered in the previous week's lecture. The suggestions below are not prescriptive; they are meant to help you build familiarity with the data and the methods progressively. You can do the early exploration individually; once your group is allocated, share and combine your work in your group's shared workspace.

Week	Method (lectured the week before)	Suggestion
2	Normal linear regression	Read in both datasets. Explore them: produce summary statistics and plots, and classify each variable by type. Try a linear model for a continuous outcome.
3	Introduction to GLMs	Think about which variables are natural responses and which are explanatory, and what kind of outcome each response is.
4	Binomial regression	Consider a binary outcome in the data and fit a model for it.
5	Residuals and diagnostics	Return to a model you have fitted and check it. What do the residuals and diagnostics tell you?
6	Poisson regression	Consider a count outcome and fit a model for it.
8	Contingency tables I	Explore relationships between categorical variables in the data.
9	Contingency tables II	Continue with the categorical analyses from the previous session.
10	Specialised models	Revisit your analyses. Are there outcomes that need an approach beyond the standard models?
11	Generalised linear mixed models	Some of the data are grouped. Think about whether accounting for this structure changes your conclusions.
12	—	Sit down as a group. Decide which three analyses you want to present and make a plan for the report.

The full project brief will be released later in the term and will specify exactly what your group is required to produce.